Project Design Phase
Problem – Solution Fit

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Team ID	Anushka Madihali, Anuja Chavan, Pranav Sutar,Shrivardhan Patil
Project Name	Heart Disease Analysis
Maximum Marks	4 Marks

Problem – Solution Fit

The Problem-Solution Fit means that a real problem has been identified among target customers, and the proposed solution actually and effectively solves that problem. It helps identify behavioral patterns and recognize what would work and why.

Purpose of Problem-Solution Fit:

- Solve complex problems in a way that fits the state of your customers.
- Succeed faster and increase solution adoption by tapping into existing behaviors and channels.
- Sharpen communication and strategy with the right triggers and messaging.
- Increase touchpoints by finding the right problem-behavior fit and building trust by solving frequent, urgent, or costly problems.
- Understand the existing situation in order to improve it for the target group.

Problem – Solution Fit Template

Element	Details
Target Customer	Cardiologists (Dr. Sharma), Government Health Policy Analysts (Ramesh), and health-conscious individuals with risk factors (Anita, 45 years old) across healthcare and public health domains.
Observed Problem	Difficulty in analyzing heart disease risk factors across 18 variables — BMI, smoking, age, gender, diabetes, physical activity, stroke — from a dataset of 4,500 records without a centralized visual tool.
Why It's a Problem	Healthcare professionals and individuals must manually browse raw CSV data and static spreadsheet reports, causing confusion, time delays, missed correlations, and uncertainty in clinical decision-making and personal health management.
Your Solution	A comprehensive Tableau dashboard showing heart disease patterns across gender, age, race, lifestyle habits, and existing health conditions. Includes 10 interactive visualizations, an assembled dashboard, and a Tableau Story titled 'Heartbeat of Data'.

Why It Works	Centralizes 18 risk factor variables using clear, interactive visuals and filters — making multi-dimensional analysis quick and accessible for both clinical professionals and non-technical individuals without requiring data science expertise.
How You Validate Fit	Tested through creation of 10 Tableau visualizations that clearly reveal patterns — males aged 55–74 at highest risk, smokers showing higher heart disease rates, diabetic patients with elevated stroke risk — proving the dashboard delivers actionable insights for real decision-making.

Detailed Problem – Solution Mapping

The table below maps each customer problem to the specific solution feature that addresses it, showing how the Heart Disease Tableau dashboard resolves each identified pain point for each stakeholder.

Problem	Customer	Goal	Pain Point	Consequence	Solution Feature
Problem 1	Cardiologist (Dr. Sharma)	Identify high-risk patients from lifestyle data quickly	No centralized visual tool exists for multi-variable patient risk analysis	Clinical decisions based on incomplete manual data review	Interactive Tableau dashboard with filters for BMI, smoking, age, and gender
Problem 2	Health Analyst (Ramesh)	Study heart disease trends across demographics for policy work	Data is spread across static reports without interactive regional analysis	Policy recommendations lack visual, evidence-based data support	Demographic and lifestyle dashboards with race, age, and activity filters
Problem 3	Individual (Anita, 45yr)	Understand personal heart disease risk compared to benchmarks	No simplified personal health dashboard from healthcare providers	Individuals cannot compare their own risk profile to healthy populations	Personalized risk visualization with general health and BMI comparisons

Validation of Problem-Solution Fit

Evidence from Visualizations

The following insights from the 10 Tableau visualizations confirm that the dashboard successfully addresses the identified problems:

- Gender vs Heart Disease chart reveals females (227 cases) and males (256 cases) have measurably different heart disease rates — directly addressing Dr. Sharma's need for demographic risk breakdowns.

- Age vs Heart Disease shows a clear peak in the 65-69 age group — providing Ramesh with evidence for age-targeted health policy interventions.
- Smoking & Alcohol impact chart shows smokers with no alcohol (139 cases) vs smokers with alcohol (123 cases) — confirming lifestyle correlation insights for prevention campaigns.
- General Health vs Heart Disease bubble chart shows 'Good' health patients (172 cases) still developing heart disease — highlighting the importance of proactive screening over self-reported health status.
- Race-wise distribution shows White population at 75.57% of heart disease cases — enabling Ramesh to prioritize outreach by demographic group.

Fit Conclusion

The Heart Disease Analysis Tableau dashboard demonstrates a strong Problem-Solution Fit. Each of the three target customer pain points — clinical risk identification, policy analytics, and personal health monitoring — is directly addressed by specific dashboard features, filters, and visualizations built on the 4,500-record patient dataset.